Honeywell

Asset Performance Management

Developer's User Guide

Release 540.1

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CHAPTER

1 ABOUT THIS GUIDE

This guide describes the uses of the Asset Performance Management (APM).

1.1 Revision history

Revision	Date	Description
1	August 2023	Asset Performance Management R540.1 General Availability (GA) release

1.2 Intended audience

This documents is intended for engineers who want to understand Asset Performance Management.

1.3 Related documents

The following list identifies publications that may contain information relevant to the information in this document.

Document Name	Document Number
Asset Performance Management User's Guide	APMDOC-X392-en-R540.1
Asset Performance Management Configuration Guide	APMDOC-X391-en-R540.1
Asset Performance Management Installation Guide	APMDOC-X390-en-R540.1
Asset Performance Management Upgrade Guide	APMDOC-X390-en-R540.1
Asset Performance Management Troubleshooting Guide	APMDOC-X393-en-R540.1
Asset Performance Management R540.1 Software Change Notice	APMDOC-X166-en-R540.1

CHAPTER

2 Introduction to the Asset Performance Management APM

The APM provides the following features which were previously accessible only within the Asset Performance Management (APM) software. These features are now available using an open set of interfaces that you can call from a .NET application.

The APM can be used for the following tasks:

1. Inspect tag values from live calculations.
2. Modify input data to adjust inputs to calculations.
3. Simulate calculations.
4. Write data points to trigger data-driven calculations.
5. Access the real-time data cache and historian.
6. Integrate APM into your own solutions.

You need to have programming knowledge to create your own application that uses the APM. Alternatively, you can use the **Calcutron 5000** command-line tool, which can call the APM methods on your behalf using an executable program.

CHAPTER

3 THE CALCUTRON 5000 TOOL

The Calcutron 5000 EXE is a simple tool that you can use to call the APM methods. Using command line parameters, you can activate the APM methods. This option is useful if you do not have Visual Studio or if you want to integrate the APM into an ETL tool.

APM must be installed prior to using this tool. It assumes that HTTPS is enabled and defaults to using **localhost**. However, you can override this setting in the command line options.

The below table contains the list of options available and you can generate this help output by invoking **calcutron --help**.

Usage:

`calcutron <options>`

Options	Description
`--write`	Write a tag to the RTX cache.
`--read`	Read a tag from the RTX cache
`--trigger`	Run a calculation in the model
`--simulate`	Load a calculation and run it.
`-h, --host=VALUE`	The host name or IP address of the Asset Performance Management server.
`-e, --equipment=VALUE`	Equipment which contains the calculation.
`-c, --calculation=VALUE`	Calculation name.
`-t, --tag=VALUE`	The tag to read or write.
`-v, --value=VALUE`	The value to write to the tag.
`-m, --model=VALUE`	Model which contains the equipment.
`-?, --help`	Show this message and exit.

Examples

Write a value to the real-time data cache for tag101:

```
calcutron -write -t tag101 -v 123.45
```

Write a value on a different host:

```
calcutron -h uasserver3.domain.com -write -t Pump01-FlowRate -v 123.45
```

Read a value from the real-time data cache:

```
calcutron -read -t Pump01-FlowRate
```

Trigger an existing calculation attached to the specified equipment and model:

```
calcutron -trigger -e pump15 -c calc5 -m model1
```

CHAPTER

4 THE ASSET PERFORMANCE MANAGEMENT APM

The following topics will assist you to set up and integrate Asset Performance Management. To get started, you need Visual Studio 2015 installed with .NET 4.71.

4.1 Dependencies

The APM uses classes from the APM core system. These client-side libraries should be included in your project so you can invoke the APM methods. You will need to reference the following assemblies in your Visual Studio 2015 project.

- References
  - Analyzers
  - EntityFramework
  - EntityFramework.SqlServer
  - Honeywell.AM.Types
  - Honeywell.AM.Utility
  - Honeywell.Common.RTDataService
  - Honeywell.MES.Core.DataModel.EntityRuntime
  - Honeywell.RTPM.Common.Types
  - Honeywell.RTPM.DataService
  - SentinelBaseDataModel
  - SentinelCoreDataModel
  - SentinelMainDataModel
  - System
  - System.ComponentModel.DataAnnotations
  - System.configuration
  - System.Core
  - System.Data
  - System.Data.DataSetExtensions
  - System.Deployment
  - System.Drawing
  - System.ServiceModel
  - System.Windows.Forms
  - System.Xml
  - System.Xml.Linq

4.2 How to Read and Write Data

Use the Utility class to open a connection to the RTx. The host name should be provided as part of the URL.

```
 IDataService ds = Utility.MakeBaseProxy<IDataService>("net.tcp://<host
name>:9022/DataService");
```

Example: Write a Data Point

```
static void RunDataPointWrite(string host, string name, string value)
{
```

```
Console.WriteLine("Writing data point: Host="+host+", Tag="+name+",
Value="+value);
IDataService ds = Utility.MakeBaseProxy<IDataService>("net.tcp://" + host +
":9022/DataService");
double dvalue = double.NaN
if (Double.TryParse(value, out dvalue))
{
RtxData data = new RtxData(name, value);
ds.WriteData(data);
Console.WriteLine("Successfully wrote value to DataService");
}
else
{
Console.WriteLine("Invalid tag value, must be double: "+value);
}
}
```

Example: Read a Data Point

```
static void RunDataPointRead(string host, string name)
{
Console.WriteLine("Reading data point: Host = "+host+", Tag = "+name);
String cx = "net.tcp://" + host + ":9022/DataService";
Console.WriteLine("Connecting to DataService: ");
IDataService ds = Utility.MakeBaseProxy<IDataService>(cx);
RtxData data = null;
Console.WriteLine("Calling ReadData() on DataService");
ds.ReadData(name, out data);
if (null == data || null == data.Value)
{
Console.WriteLine("Could not find value for tag: " + name);
}
else
```

```
{
Console.WriteLine("ReadData returned: "+data);
if (data.ValueType == RtxValueType.Tabular)
{
var tabularData = data.Value as TabularData;
if (tabularData != null)
{
var table = BuildDataTable(data.Name, tabularData);
Console.WriteLine(table);
}
}
Console.WriteLine(data.Value.ToString());
}
}
```

DataService provides the following methods:

```
void Ping();
void ReadData(string key, out RtxData value);
object ReadDataObject(string key);
List<object> ReadDataObjects(List<string> keys);
void ReadDataSet(List<string> key, out List<RtxData> value);
void Remove(string key, string dataStoreName = null);
void RemoveData(string key);
void WriteData(RtxData value);
void WriteDataObject(string key, object value);
void WriteDataObjects(List<string> keys, List<object> values);
void WriteDataSet(List<RtxData> key);
void WriteDataAndNotify(RtxData value, bool ignoreDirtyCheck);
List<RtxData> WriteDataSetAndNotify(List<RtxData> incomingData, bool
ignoreDirtyCheck);
string GetConnectionTarget(string connectionAlias);
```

4.3 How to Trigger a Calculation

Calculations are central components of the Asset Performance Management system. Calculations consume input data and generate new data that can be used to model industrial processes. The APM allows us to trigger these calculations and see the results.

The central class for triggering calculations is `IRtxExecutionService`. It accepts an `ExecutionRequest` which contains the parameters that are required to find the calculation and run it.

Example: **Trigger a Calculation**

```
static void RunTriggerCalc(string host, string equipment, string calculation,
string model)
{
Console.WriteLine("Trigger calculation: Host = " + host + ", Equipment = " +
equipment + ", Calculation = " + calculation + ", Model = " + model);
// Send the request
try
{
Utility.UseHttpProxy<IRtxExecutionService>(
"https://" + host + "/MES/AM/ExecutionService/Execution.svc",
(service) =>
{
service.ExecuteModel(new ExecutionRequest()
{
ExecutionMode = ExecutionMode.Trigger,
ModelLoadLocation = ModelLoadLocation.LiveCache,
EquipmentCode = equipment,
ModelNames = new List<string> { model },
});
});
Console.WriteLine("Calculation trigger completed");
}
catch (Exception ex)
{
Console.WriteLine($@"Error running calc: {ex.Message}{ex.StackTrace}");
}
}
```

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